
```

n = 4;
A = rand(n, n);
x_exact = ones(n, 1);
b = A * x_exact;
[L, U, P] = descompune_LUP(A);
x_calculat = rezolva_LUP(L, U, P, b);
disp(x_calculat);

function [L_mat, U_mat, vec_perm] = descompune_LUP(Mat)
    % desc LUP pe matrice
    ordin = size(Mat, 1);
    U_mat = Mat;
    vec_perm = (1:ordin)';

    for pas = 1:ordin-1

        [~, poz_max] = max(abs(U_mat(pas:ordin, pas)));
        rand_max = pas + poz_max - 1;

        if rand_max ~= pas
            % interschimbarea
            U_mat([pas, rand_max], :) = U_mat([rand_max, pas], :);
            % actualizarea vect
            vec_perm([pas, rand_max]) = vec_perm([rand_max, pas]);
        end

        % actualizarea si la matrice
        for rand_urm = pas+1:ordin
            U_mat(rand_urm, pas) = U_mat(rand_urm, pas) / U_mat(pas, pas);
            U_mat(rand_urm, pas+1:ordin) = U_mat(rand_urm, pas+1:ordin) -
U_mat(rand_urm, pas) * U_mat(pas, pas+1:ordin);
        end
    end

    % triunghiulara inferioara si superioara
    L_mat = tril(U_mat, -1) + eye(ordin);
    U_mat = triu(U_mat);
end

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1.0000
1.0000
1.0000

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